

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Correlation and Pattern Analysis

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO4: Explore datasets using analytical techniques such as grouping, correlation detection, joining datasets, and extracting meaningful insights.

Assessment Tool Weightage: 25%

Total Marks: 20

Code of Conduct

I **Naresh M(192424381)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Naresh M

Correlation and Pattern Analysis Rubric

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Data Importing, Exploration & Preparation	25	Thorough understanding; correctly imports, explores, cleans, and prepares data;	Good understanding; performs most data preparation tasks correctly with minor issues	Basic understanding; performs some preparation tasks but misses important issues	Poor understanding; unable to import, explore, or prepare the data correctly
Python Coding & Data Analysis	25	Effectively applies relevant Python concepts; code is accurate, efficient, and performs analysis appropriately	Applies appropriate Python concepts with minor coding or analytical gaps	Limited application of Python concepts; analysis is partially completed	Fails to apply relevant Python concepts; major errors or incomplete analysis
Correlation Analysis & Visualization	20	Performs in-depth correlation analysis with appropriate visualizations; demonstrates strong reasoning and insights	Performs good correlation analysis with suitable visualizations and logical reasoning	Performs basic correlation analysis with limited visualization or interpretation	Weak or no correlation analysis; inappropriate/missing visualizations and lacks reasoning
Pattern Identification &	20	Identifies meaningful and insightful patterns; provides	Identifies relevant patterns and provides logical	Identifies basic patterns with weak	Incorrect or no meaningful patterns identified;

Interpretation		innovative, feasible interpretations with strong justification	interpretations with adequate justification	or limited justification	interpretation is missing or inappropriate
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

Total Marks Awarded:

Signature of the Course Faculty

Questions:

5. E-Commerce Dataset: Discovering Hidden Purchase Patterns

An e-commerce company has three datasets:

customers.csv

Customer_ID, Age, Gender, City

C01, 21, M, Chennai

C02, 35, F, Bangalore

C03, 29, M, Chennai

C04, 45, F, Hyderabad

C05, 38, M, Bangalore

orders.csv

Order_ID, Customer_ID, Product_Category, Quantity, Order_Value

O01, C01, Electronics, 2, 2500

O02, C02, Clothing, 4, 3200

O03, C03, Electronics, 1, 1800

O04, C04, Furniture, 2, 7000

O05, C05, Clothing, 5, 4500

O06, C01, Clothing, 3, 2100

O07, C02, Electronics, 2, 3000

O08, C03, Clothing, 2, 1600

Task: Write a Python program to perform the following analysis:

1. Import all datasets using Pandas.
2. Explore each dataset using appropriate DataFrame functions.
3. Join the customer and order datasets using Customer_ID.
4. Calculate correlations between:
 - Age and Order Value
 - Quantity and Order Value
5. Group the orders by Product_Category and calculate:
 - Total quantity sold
 - Total order value
 - Average order value
6. Identify the product category with the highest total sales.
7. Create suitable visualizations to identify purchasing patterns.
8. Analyze whether quantity ordered is strongly related to order value.
9. Identify at least three meaningful patterns from the combined dataset.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from io import StringIO

customers_data = """Customer_ID,Age,Gender,City
C01,21,M,Chennai
C02,35,F,Bangalore
C03,29,M,Chennai
C04,45,F,Hyderabad
C05,38,M,Bangalore"""

orders_data = """Order_ID,Customer_ID,Product_Category,Quantity,Order_Value
O01,C01,Electronics,2,2500
O02,C02,Clothing,4,3200
O03,C03,Electronics,1,1800
O04,C04,Furniture,2,7000
```

O05,C05,Clothing,5,4500

O06,C01,Clothing,3,2100

O07,C02,Electronics,2,3000

O08,C03,Clothing,2,1600""

```
customers = pd.read_csv(StringIO(customers_data))
```

```
orders = pd.read_csv(StringIO(orders_data))
```

```
print("Customer Dataset:")
```

```
print(customers)
```

```
print("\nOrder Dataset:")
```

```
print(orders)
```

```
print("\nCustomer Data Types:")
```

```
print(customers.dtypes)
```

```
print("\nOrder Data Types:")
```

```
print(orders.dtypes)
```

```
print("\nCustomer Summary:")
```

```
print(customers.describe())
```

```
print("\nOrder Summary:")
```

```
print(orders.describe())
```

```
combined = pd.merge(customers, orders, on="Customer_ID")
```

```
print("\nCombined Dataset:")
```

```
print(combined)
```

```
age_corr = combined["Age"].corr(combined["Order_Value"])
```

```
quantity_corr = combined["Quantity"].corr(combined["Order_Value"])
```

```
print("\nCorrelation between Age and Order Value:", round(age_corr, 3))
```

```
print("Correlation between Quantity and Order Value:", round(quantity_corr, 3))
```

```
category_summary = orders.groupby("Product_Category").agg(
```

```
    Total_Quantity=("Quantity", "sum"),
```

```
    Total_Order_Value=("Order_Value", "sum"),
```

```
    Average_Order_Value=("Order_Value", "mean")
```

```
)
```

```
print("\nProduct Category Analysis:")
```

```

print(category_summary)

highest_sales = category_summary["Total_Order_Value"].idxmax()

print("\nProduct Category with Highest Total Sales:")

print(highest_sales)

plt.figure(figsize=(8, 5))

sns.barplot(
    x=category_summary.index,
    y=category_summary["Total_Order_Value"]
)

plt.title("Total Sales by Product Category")

plt.xlabel("Product Category")

plt.ylabel("Total Order Value")

plt.show()

plt.figure(figsize=(7, 5))

sns.scatterplot(
    data=combined,
    x="Quantity",
    y="Order_Value",
    hue="Product_Category",
    s=100
)

plt.title("Quantity vs Order Value")

plt.xlabel("Quantity")

plt.ylabel("Order Value")

plt.show()

plt.figure(figsize=(7, 5))

sns.heatmap(
    combined[["Age", "Quantity", "Order_Value"]].corr(),
    annot=True,
    cmap="coolwarm",
    fmt=".3f"
)

```

```
)  
plt.title("Correlation Heatmap")  
plt.show()
```

Output:

```
===== RESTART: C:/Users/nares/Videos/dsa0503/experiments/ass 4.2.py =====
```

```
Customer Dataset:
```

	Customer_ID	Age	Gender	City
0	C01	21	M	Chennai
1	C02	35	F	Bangalore
2	C03	29	M	Chennai
3	C04	45	F	Hyderabad
4	C05	38	M	Bangalore

```
Order Dataset:
```

	Order_ID	Customer_ID	Product_Category	Quantity	Order_Value
0	O01	C01	Electronics	2	2500
1	O02	C02	Clothing	4	3200
2	O03	C03	Electronics	1	1800
3	O04	C04	Furniture	2	7000
4	O05	C05	Clothing	5	4500
5	O06	C01	Clothing	3	2100
6	O07	C02	Electronics	2	3000
7	O08	C03	Clothing	2	1600

```
Customer Data Types:
```

```
Customer_ID      str  
Age              int64  
Gender           str  
City            str  
dtype: object
```

```
Order Data Types:
```

```
Order_ID         str  
Customer_ID      str  
Product_Category str  
Quantity         int64  
Order_Value      int64  
dtype: object
```

Customer Summary:

	Age
count	5.000000
mean	33.600000
std	9.099451
min	21.000000
25%	29.000000
50%	35.000000
75%	38.000000
max	45.000000

Order Summary:

	Quantity	Order_Value
count	8.000000	8.000000
mean	2.62500	3212.500000
std	1.30247	1788.405115
min	1.00000	1600.000000
25%	2.00000	2025.000000
50%	2.00000	2750.000000
75%	3.25000	3525.000000
max	5.00000	7000.000000

Combined Dataset:

	Customer_ID	Age	Gender	...	Product_Category	Quantity	Order_Value
0	C01	21	M	...	Electronics	2	2500
1	C01	21	M	...	Clothing	3	2100
2	C02	35	F	...	Clothing	4	3200
3	C02	35	F	...	Electronics	2	3000
4	C03	29	M	...	Electronics	1	1800
5	C03	29	M	...	Clothing	2	1600
6	C04	45	F	...	Furniture	2	7000
7	C05	38	M	...	Clothing	5	4500

[8 rows x 8 columns]

Correlation between Age and Order Value: 0.823

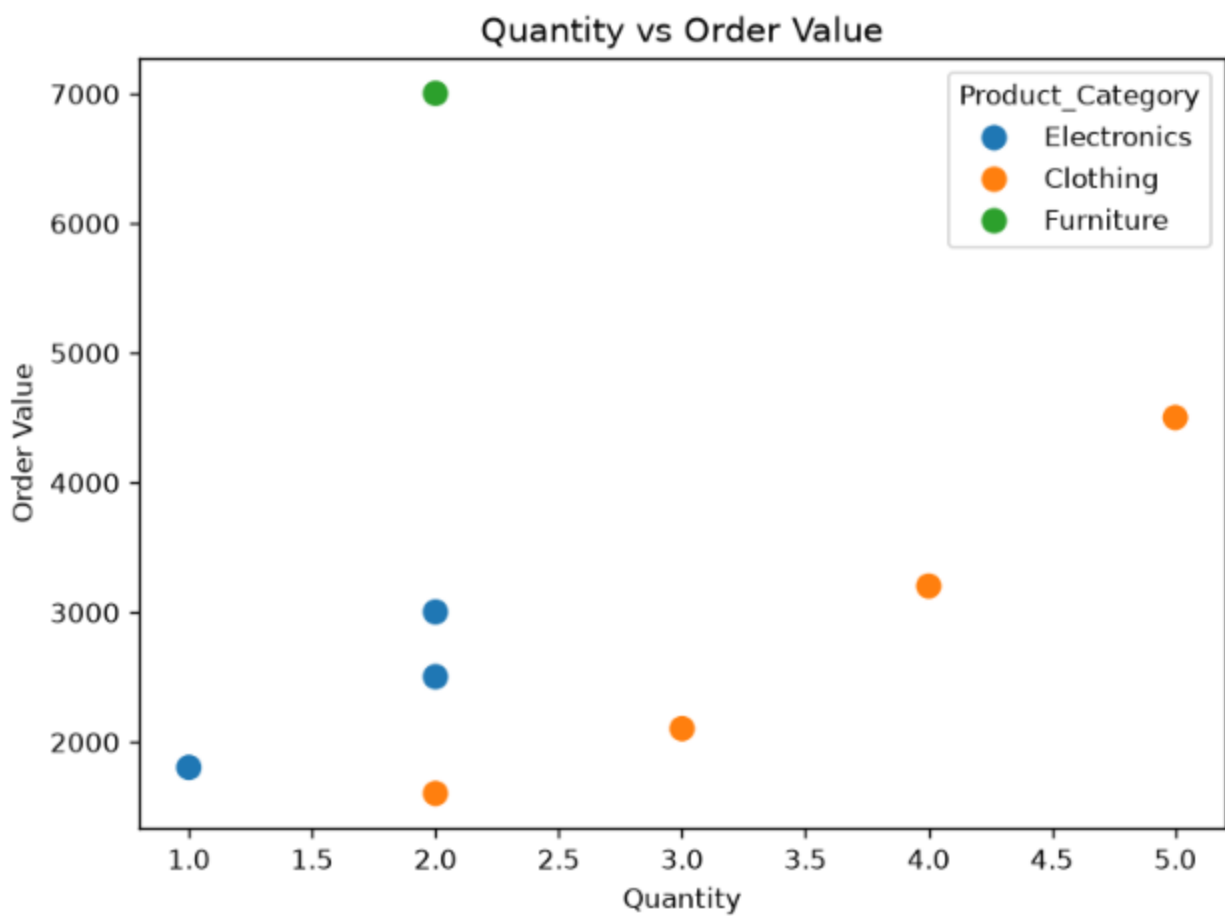
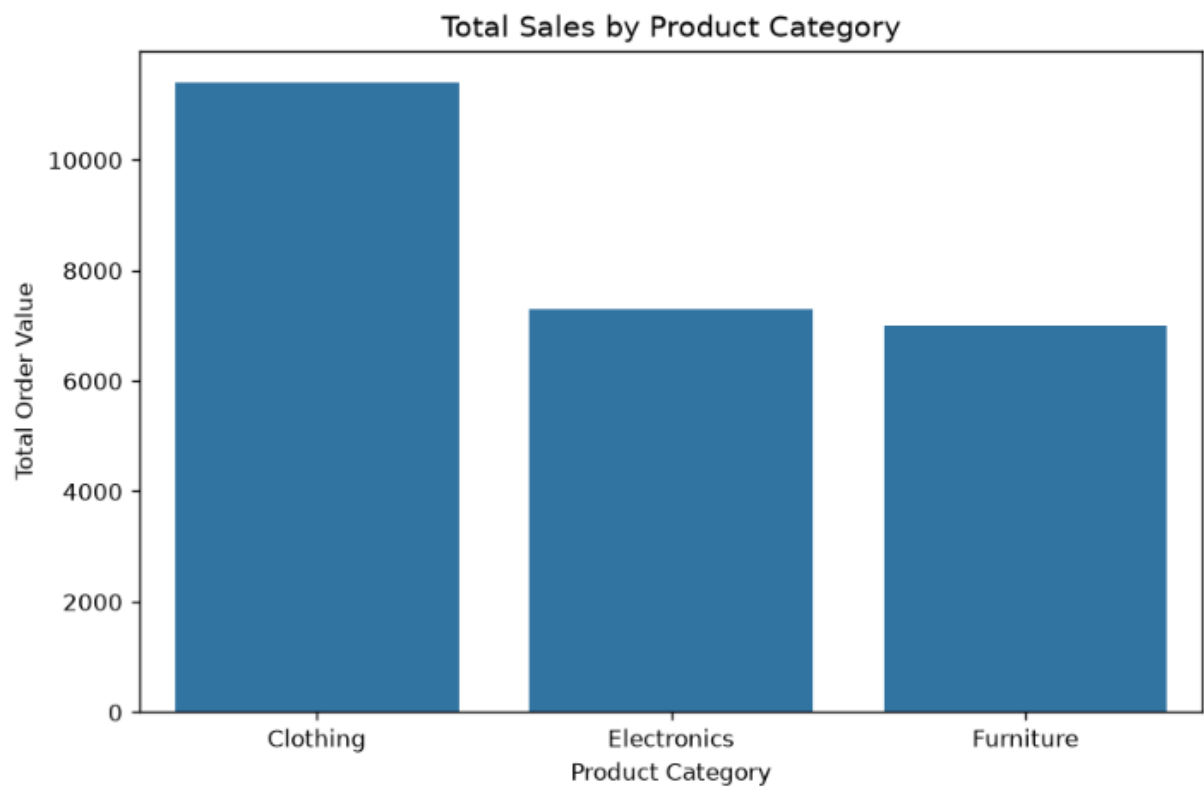
Correlation between Quantity and Order Value: 0.254

Product Category Analysis:

	Total_Quantity	Total_Order_Value	Average_Order_Value
Product_Category			
Clothing	14	11400	2850.000000
Electronics	5	7300	2433.333333
Furniture	2	7000	7000.000000

Product Category with Highest Total Sales:

Clothing



2. Correlation Results

Relationship	Correlation
Age vs Order Value	0.823
Quantity vs Order Value	0.254

Interpretation:

- **Age and Order Value = 0.823** → strong positive relationship.
- **Quantity and Order Value = 0.254** → weak positive relationship.

Age has a much stronger relationship with Order Value than Quantity does.

3. Product Category Analysis

Product Category	Total Quantity	Total Order Value	Average Order Value
Clothing	14	11,400	2,850.00
Electronics	5	7,300	2,433.33
Furniture	2	7,000	7,000.00

4. Highest Total Sales

Clothing has the highest total sales.

- Total quantity sold = **14**
- Total order value = **₹11,400**
- Average order value = **₹2,850**

Although Furniture has only 2 units sold, its average order value is **₹7,000**, which is much higher than Clothing and Electronics.

5. Three Meaningful Patterns

Pattern 1 – Age and spending:

Age has a strong positive correlation (0.823) with Order Value. In this dataset, older customers generally tend to place higher-value orders.

Pattern 2 – Quantity and spending:

Quantity has only a weak positive correlation (0.254) with Order Value. Therefore, ordering more items does not necessarily mean a proportionally higher order value.

Pattern 3 – Product category:

Clothing generates the highest total sales (₹11,400) and has the highest total quantity sold (14 units). Furniture has the highest average order value (₹7,000) despite having only two units sold.

6. Conclusion

The analysis shows that **customer age has a stronger relationship with order value than quantity ordered**. Clothing is the most popular category in terms of total quantity and total sales, while Furniture generates the highest average order value. Therefore, the e-commerce business could focus on **high-value products such as Furniture** while maintaining the strong sales volume of **Clothing**.